SEMESTER EXAMINATION- DECEMBER 2022
CLASS: SEMESTER:
PAPER CODE: BEM-C302
PAPER TITLE: ENGINEERING MATHEMATICS III

Time: 3 hour Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

Instructions: Answer any ***five*** questions in about 150 words each. Each question carries six marks.
(5 X 6 = 30 Marks)

Question-1: Find Laplace transform of $t.\int_0^t e^{-2t} \sin 3t dt$.

Question-2: Evaluate $\int_0^\infty \frac{e^{-t}\sin t}{t} dt$ and $\int_0^\infty \frac{\sin t}{t} dt$

Question-3: Use Convolution theorem to find inverse Laplace transform of $\frac{1}{(s-3)(s+3)^2}$.

Question-4: Find the Fourier transform of $f(x) = \begin{cases} 1-|x| & if\ |x| < 1 \\ 0 & |x| > 1 \end{cases}$
hence evaluate $\int_0^\infty \left(\frac{\sin t}{t}\right)^4 dt$.

Question-5: Express the function $f(x) = \begin{cases} 1 & 0 \le x < \pi \\ 0 & x > \pi \end{cases}$
as a Fourier sine integral and hence evaluate $\int_0^\infty \frac{1-\cos \pi\lambda}{\lambda} \sin\lambda x d\lambda$.

Question-6: Find inverse Z transform of $\frac{z^2}{(z-2)(z-3)}$ for $2 < |z| < 3$.

Question-7: If $F(z) = \frac{2z^2+3z+14}{(z-1)^4}$ find f_2 and f_3.

Question-8: Show that the function $u = x^3 - 3xy^2$ is harmonic find its harmonic conjugate v and corresponding analytic function f(z).

Question-9: Evaluate $\oint_C \frac{z+1}{z(z-2)} dz$ where (i) $C: |z+3| = 1$ (ii) $C: |z| = 2$

Question-10: Use Newton Raphson method to find the real root of $3x = \cos x + 1$ correct to 4 decimal places.

SECTION-B (LONG ANSWER TYPE QUESTIONS)

Instructions: Answer any ***four*** questions in detail. Each question carries 10 marks.

(4 X 10 = 40 Marks)

Question-1: Use Laplace Transform to solve $y'' + 2y' - 3y = 6e^{-2t}$ given $y(0) = 2, y'(0) = 1$

Question-2: Solve $\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$ for $0 \le x < \infty$ given the conditions

$(i) u(x,0) = 0 \text{ for } x \ge 0 \quad (ii) \frac{\partial u}{\partial x}(0,t) = -2 \quad (iii)\ u(x,t) \text{ is bounded.}$

Question-3: Find the Fourier transform of $f(x) = \begin{cases} 1 - x^2 & |x| \le 1 \\ 0 & |x| > 1 \end{cases}$

Hence evaluate $\int_0^{\infty} \frac{\sin x - x\cos x}{x^3} \cos\frac{x}{2}\, dx$.

Question-4: Use Z transform to solve $y_{n+2} - 5y_{n+1} + 6y_n = 1, \quad given\ y_0 = 0, y_1 = 1.$

Question-5: Evaluate the integral $\int_C |z| dz$, where C is the contour

(i) The straight line from $z = -i$ to $z = i$.

(ii) The left half of the unit circle $|z| = 1$ from $z = -i$ to $z = i$.

Question-6: Use Contour integration to evaluate $\int_0^{2\pi} \frac{d\theta}{5 - 3\cos\theta}$.

Question-7: Find the real root of the equation $3x + \sin x - e^x = 0$ by secant method correct to four decimal places if the root lies between (0, 1).

Question-8: Use bisection method to approximate the real root of $x^3 - x - 1 = 0$ correct to 3 decimal places if the root lies between (1, 2).